Aircraft Design
AE 405

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Team - Term Project

Grading

▪ Conceptual design of fighter aircraft,	▪ 40 points
▪ Solid modeling of aircraft Shell,	▪ 20 points
▪ Production of aircraft (wood or foam),	▪ 20 points
▪ Flight test of aircraft (r/c radio controlled)	▪ 20 points

If it flies successfully +20 points

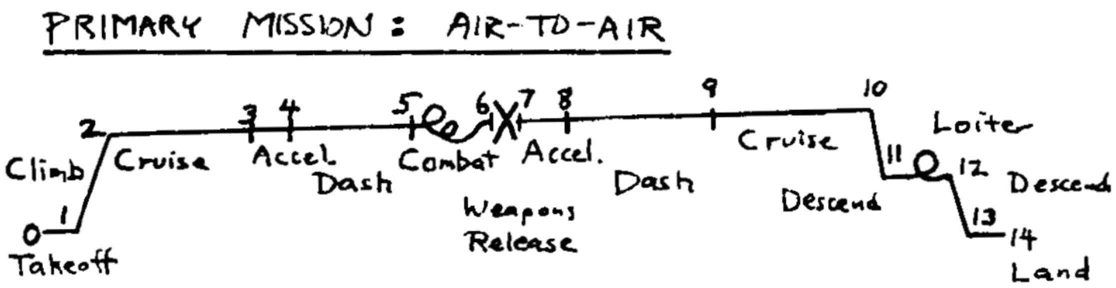
Due date: A week ahead of final exam.

Term Project

Design Requirements

- Unmanned,
- Double-engine (new, ‘rubber’),
- Stealth, Supercruise,
- Primary mission: air to air combat.

Mission profile



- 3 & 10 (cruise): 250 nm at best cruise Mach and altitude,
- 5 & 9 (dash): 75 nm at 1.5 M at 30,000 ft.
- 6 (Combat): 4 min at max. thrust, M 0.9 at 30,000 ft,
- 12 (Loiter): 30 min at sea level, best loiter speed,
- 7 (weapons release): 400 lb (missiles only)

Term Project

- Payload:
  - 1 laser gun,
  - 2 advanced missiles (200 lb, 5 in x 90 in)
  - Advanced gun (400 lb), 750 rounds ammo (450 lb)
  - Pilot (220 lb)

- Performance requirements;
  - Take off and landing (1000 ft ground roll),
  - Approach speed $\leq$ 120 kts,
  - Maximum Mach $\geq$1.8 (A/B); M 1.5 (Dry),
  - Accelerate M 0.9 to M 1.5 in 30 sec at 30,000 ft,
  - Ps =0 at 5g at 25,000 ft at M 0.9 and M 1.5,
  - $\dot{\varphi} \geq 20\ {}^{\circ}/_{sec}$ at 350 kts at 25,000 ft.

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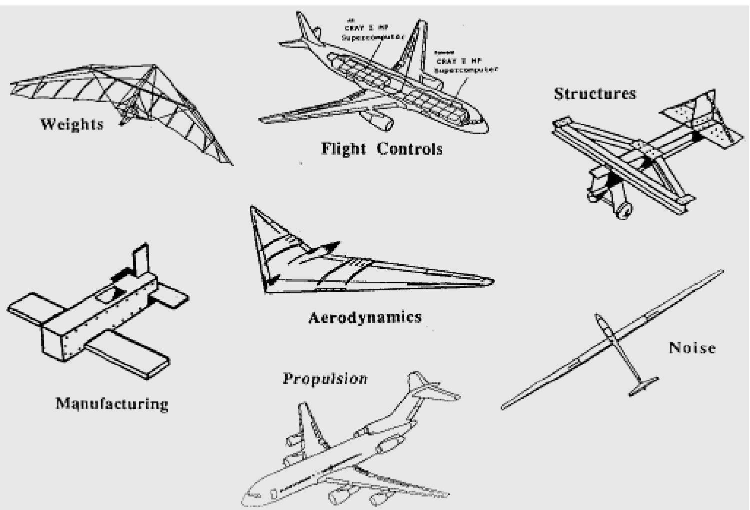
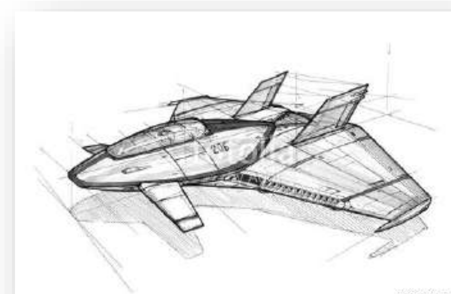


Chapter 1 - 2 - 3

- Design – a separate discipline
- Overview of the design process
- Sizing from a conceptual sketch

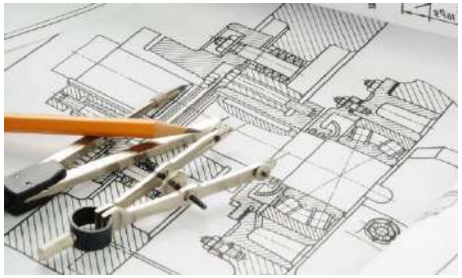


Design – a separate discipline

- Aircraft design is a separate discipline of aeronautical engineering different from the analytical disciplines such as aerodynamics, structures, controls, and propulsion.

- An aircraft designer spends time doing something called "design," creating the geometric description of a thing to be built.

- "Design" looks a lot like "drafting", or in the modern world, "computer-aided drafting".



Design – a separate discipline

A good aircraft design looks like…

F -22

Su -34

- Somehow, the landing gear fits,
- The fuel tanks are near the center of gravity,
- The structural members are simple and lightweight,
- The overall arrangement provides good aerodynamics,
- The engines install in a simple and clean fashion,
- And a host of similar detail seems to fall into place…

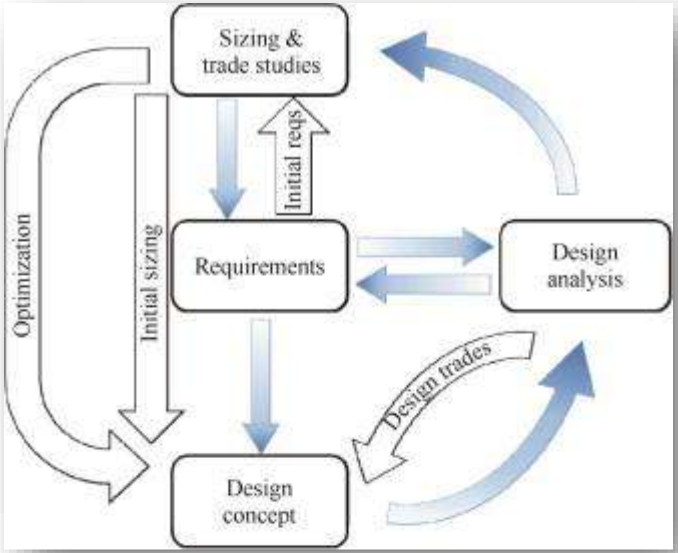
Overview of the design process

- Design is an iterative effort, as shown in the "Design Wheel"

- Requirements are set by prior design trade studies.

- Concepts are developed to meet requirements.

- Design analysis frequently points toward new concepts and technologies, which can initiate a whole new design effort.



Overview of the design process

- Aircraft design can be broken into three major phases;
  - Conceptual design
  - Preliminary design
  - Detail Design

- Conceptual design is the primary focus of this course.

- It is in conceptual design that the basic questions of configuration arrangement size and weight, and performance are answered.

- The first question is, "Can an affordable aircraft be built that meets the requirements?" If not, the customer may wish to relax the requirements.

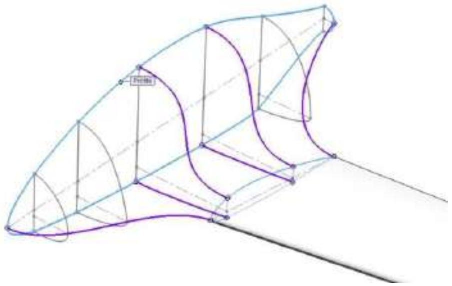
- Conceptual design is a very fluid process. New ideas and problems emerge as a design is investigated in ever-increasing detail.

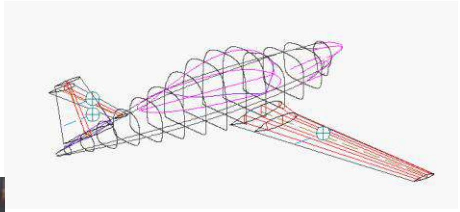
Overview of the design process

- Preliminary design can be said to begin when the major changes are over.
- The big questions such as whether to use a canard or an aft tail have been resolved.
- Testing is initiated in areas such as aerodynamics, propulsion, structures, and stability and control.
- The ultimate objective during preliminary design is to ready the company for the detail design stage, also called full-scale development.

Overview of the design process

- A key activity during preliminary design is "lofting."
- Lofting is the mathematical modeling of the outside skin of the aircraft with sufficient accuracy.
- A mockup may be constructed at this point.

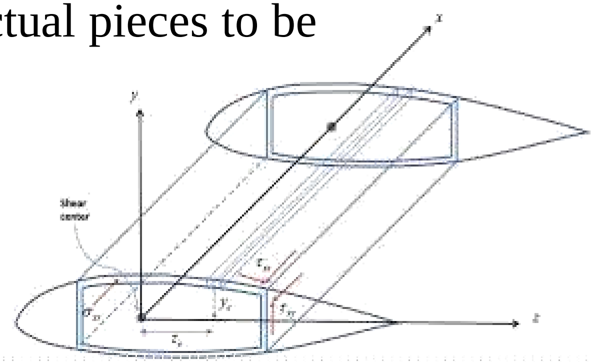




Lockheed Martin unveiled a mockup of F-35 Lighting II in Japan International Aerospace Exhibition 2016.

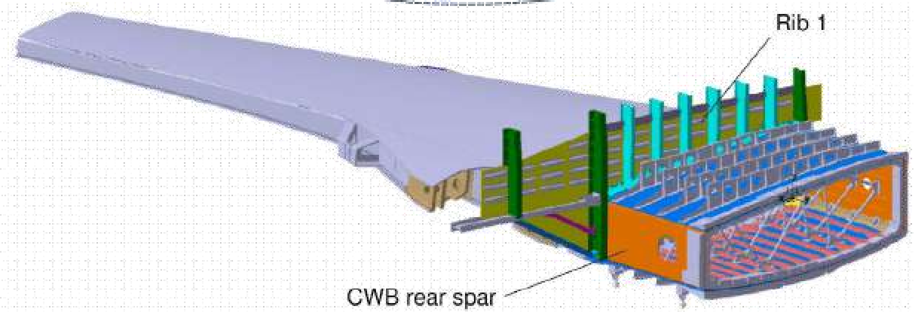
Overview of the design process

- The detail design phase begins in which the actual pieces to be fabricated are designed.



- For example, during conceptual and preliminary design the wing box will be designed and analyzed as a whole.

- During detail design, that whole will be broken down into individual ribs, spars, and skins, each of which must be separately designed and analyzed.



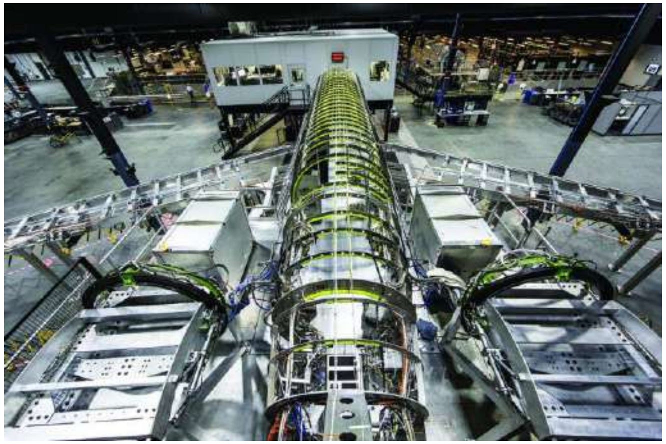
- Another important part of detail design is called production design.

- Specialists determine how the airplane will be fabricated, starting with the smallest and simplest subassemblies and building up to the final assembly process.

Overview of the design process

- During detail design, the testing effort intensifies.
  - Actual structure of the aircraft is fabricated and tested.
  - Control laws for the flight control system are tested on an "iron-bird" simulator, a detailed working model of the actuators and flight control surfaces.
  - Flight simulators are developed and flown by both company and customer test-pilots.
- Detail design ends with fabrication of the aircraft.

Gulfstream's virtual test plane, or "iron bird."

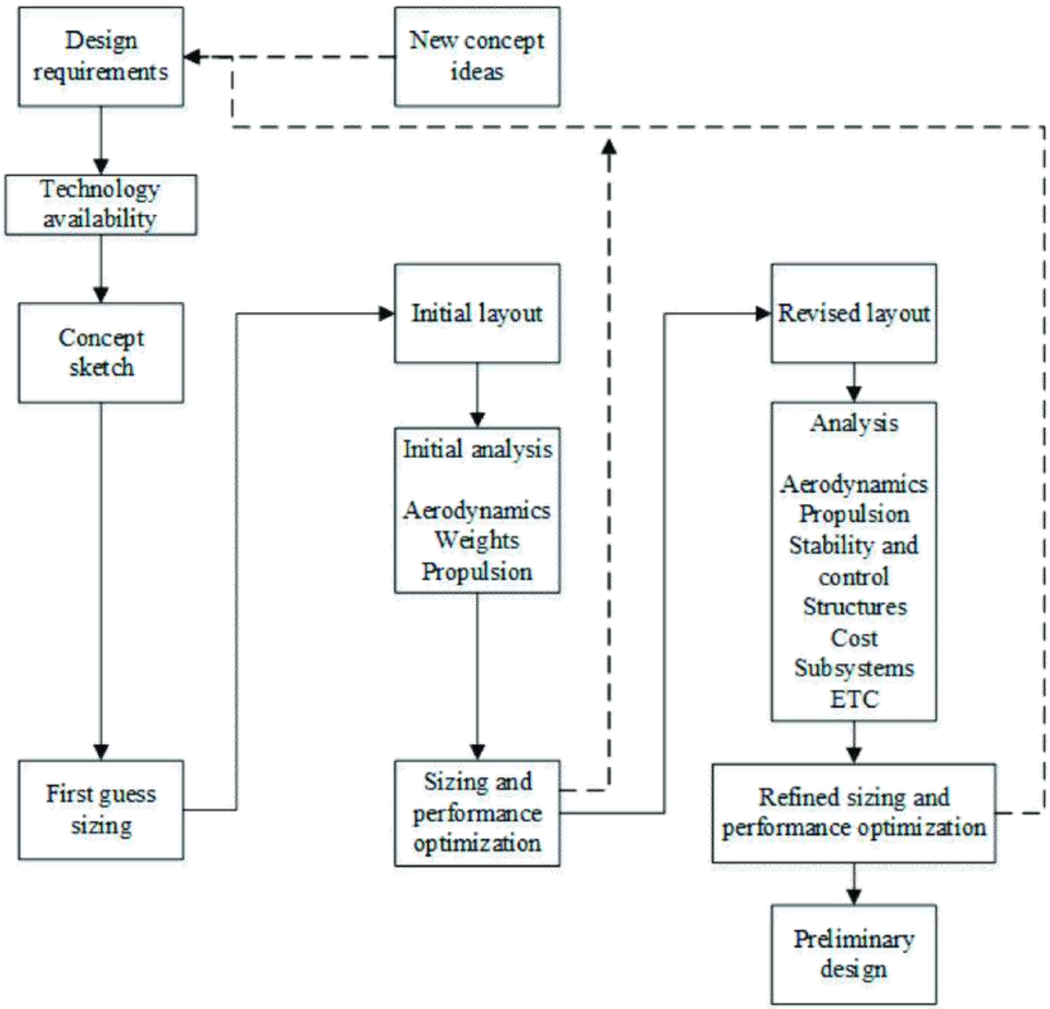


Overview of the design process

- Figure depicts the conceptual design process in greater detail.

- Conceptual design will usually begin with a set of design requirements established by the prospective customer or a company-generated guess as to what future customers may need.

- Design requirements include parameters such as
    - the aircraft range and payload,
    - takeoff and landing distances,
    - Speed requirements,
    - structural design limits,
    - pilots' outside vision angles,
    - reserve fuel, and many others



Overview of the design process

- Before a design can be started, a decision must be made as to what technologies will be incorporated.

- If a design is to be built in the near future, it must use only currently-available technologies.

- Future Technologies may result in a higher development risk.
  - Active flow control by blowing/suction pumps,
  - Laser weapons,
  - All composite fighter,
  - Blended wing body…

- The actual design effort usually begins with a conceptual sketch.

- This is the "back of a napkin" drawing of aerospace legend, and gives a rough indication of what the design may look like.

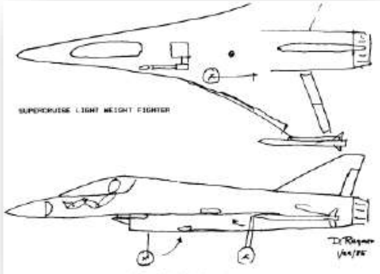


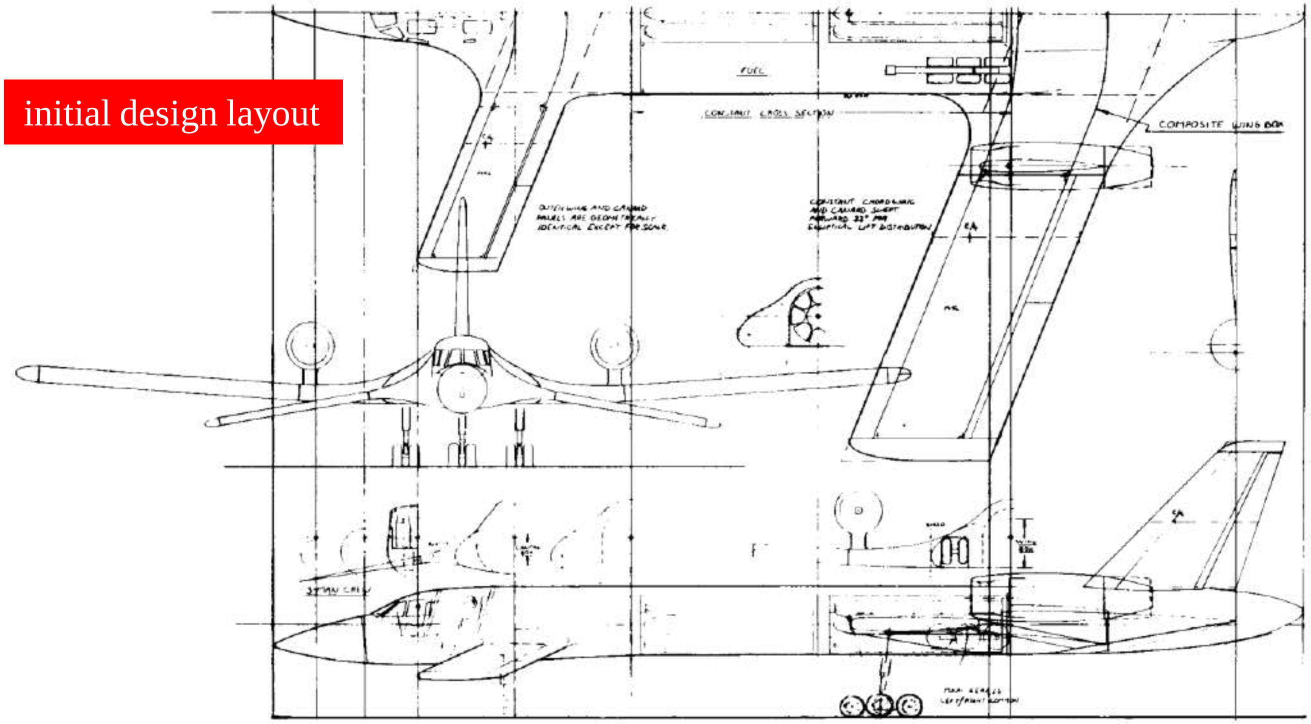
Fig. 2.4 Initial sketch.

Overview of the design process

- A good conceptual sketch will include
  - the approximate wing and tail geometries,
  - the fuselage shape, and the internal locations of the major components such as the engine, cockpit, payload/passenger compartment, landing gear, and perhaps the fuel tanks.
- The conceptual sketch can be used to estimate aerodynamics and weight fractions by comparison to previous designs.
- These estimates are used to make a first estimate of the required total weight and fuel weight to perform the design mission, by a process called "sizing."
- The "first-order" sizing provides the information needed to develop an initial design layout.

Overview of the design process

initial design layout



- On a computer-aided design system, the design work is usually done in full scale (numerically).

Overview of the design process

- The revised drawing, after some number of iterations, is then examined in detail by an ever-expanding group of specialists.

- For example, controls experts will perform a six-degree-of-freedom analysis to ensure that the designer's estimate for the size of the control surfaces is adequate for control.

- If not, they will instruct the designer as to how much each control surface must be expanded.

- The end product of all this will be an aircraft design that can be confidently passed to the preliminary design phase,

Sizing from a conceptual sketch

- There are many levels of design procedure. The simplest level just adopts past history - **statistics**.

- To get the "right" answer takes several years, many people, and lots of money.

- The most important data for sizing is the weight, especially "takeoff weight. "

- "Design takeoff gross weight" is the total weight of the aircraft as it begins the mission for which it was designed.

$$"W_0"$$

Sizing from a conceptual sketch

takeoff weight buildup

- Design takeoff gross weight can be broken into crew weight, payload (or passenger weight, fuel weight, and the remaining (or "empty") weight.
- The empty weight includes the structure, engines, landing gear, fixed equipment, avionics, and anything else not considered a part of crew, payload, or fuel.
- Equation (3.1) summarizes the takeoff-weight buildup.

$$W_0 = W_{\text{crew}} + W_{\text{payload}} + W_{\text{fuel}} + W_{\text{empty}} \tag{3.1}$$

- The crew and payload weights are both known since they are given in the design requirements.
- The only unknowns are the fuel weight and empty weight, they are both dependent on the total aircraft weight. Thus an iterative process must be used for aircraft sizing.

Sizing from a conceptual sketch

- To simplify the calculation, both fuel and empty weights can be expressed as fractions of the total takeoff weight,

$$W_0 = W_{crew} + W_{payload} + \left(\frac{W_f}{W_0}\right)W_0 + \left(\frac{W_e}{W_0}\right)W_0 \tag{3.2}$$

and

$$W_0 - \left(\frac{W_f}{W_0}\right)W_0 - \left(\frac{W_e}{W_0}\right)W_0 = W_{crew} + W_{payload}$$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - (W_f/W_0) - (W_e/W_0)}$$

- Now *Wo* can be determined if $\left(\frac{W_f}{W_0}\right)$ and $\left(\frac{W_e}{W_0}\right)$ can be estimated.

Sizing from a conceptual sketch

The empty-weight fraction

$$+\left(\frac{W_e}{W_0}\right)$$

- The empty-weight fraction can be estimated statistically from historical trends as shown in Fig. 3.1.
- Empty-weight fractions vary from about 0.3 to 0.7, and diminish with increasing total aircraft weight.
- Table 3.1 presents statistical curve-fit equations for the trends shown in Fig. 3.1.
- Note that these are all exponential equations based upon takeoff gross weight.

Sizing from a conceptual sketch

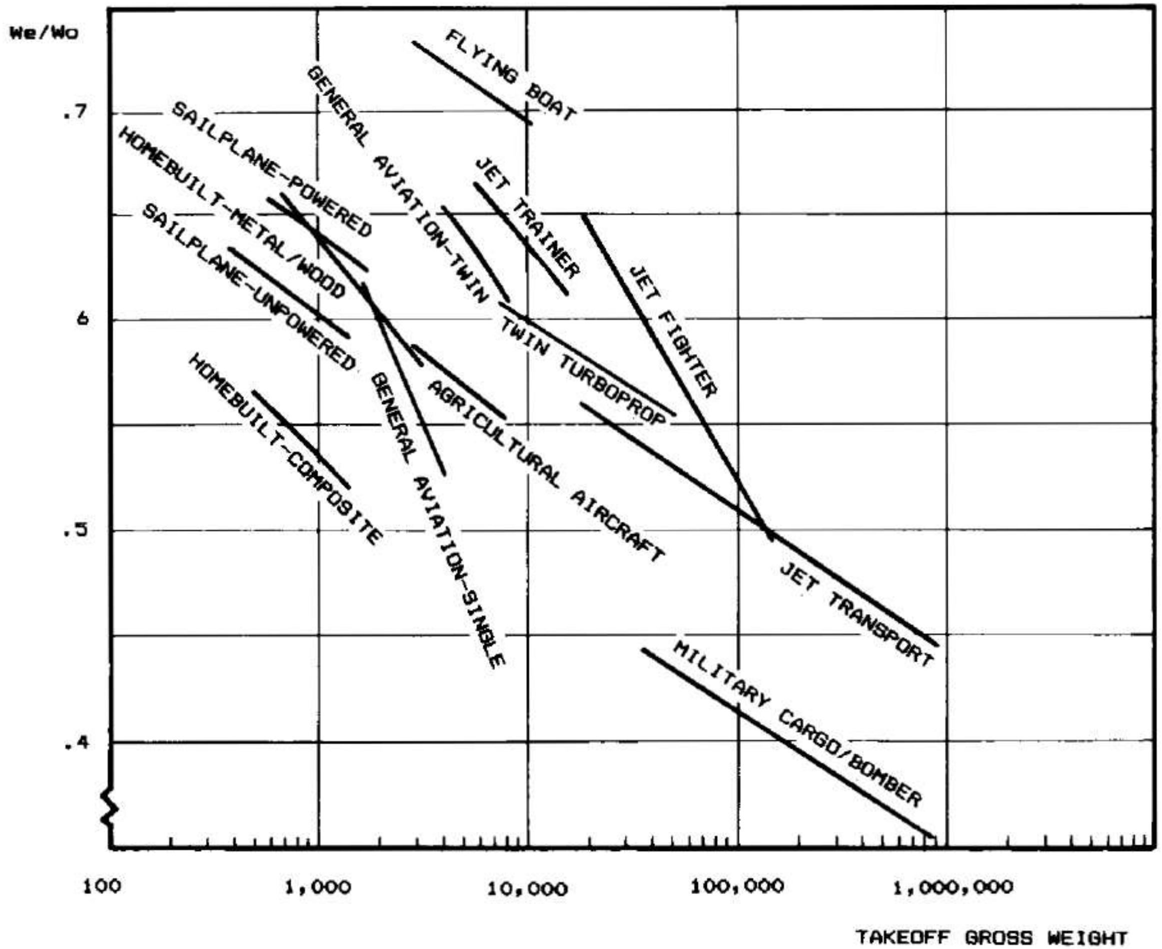


Fig. 3.1 Empty weight fraction trends.

Sizing from a conceptual sketch

Table 3.1 Empty weight fraction vs W_0

$W_e/W_0 = A W_0^C K_{vs}$	A	C
Sailplane—unpowered	0.86	−0.05
Sailplane—powered	0.91	−0.05
Homebuilt—metal/wood	1.19	−0.09
Homebuilt—composite	0.99	−0.09
General aviation—single engine	2.36	−0.18
General aviation—twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Sizing from a conceptual sketch

- Based on a number of design studies, the empty-weight fraction for composite aircraft can be estimated by multiplying the statistical empty weight fraction by 0.95.
- A variable-sweep wing is heavier than a fixed wing.
- It is accounted for at this initial stage of design by multiplying the empty-weight fraction as determined from the equations in Table 3 .1 by about 1.04.

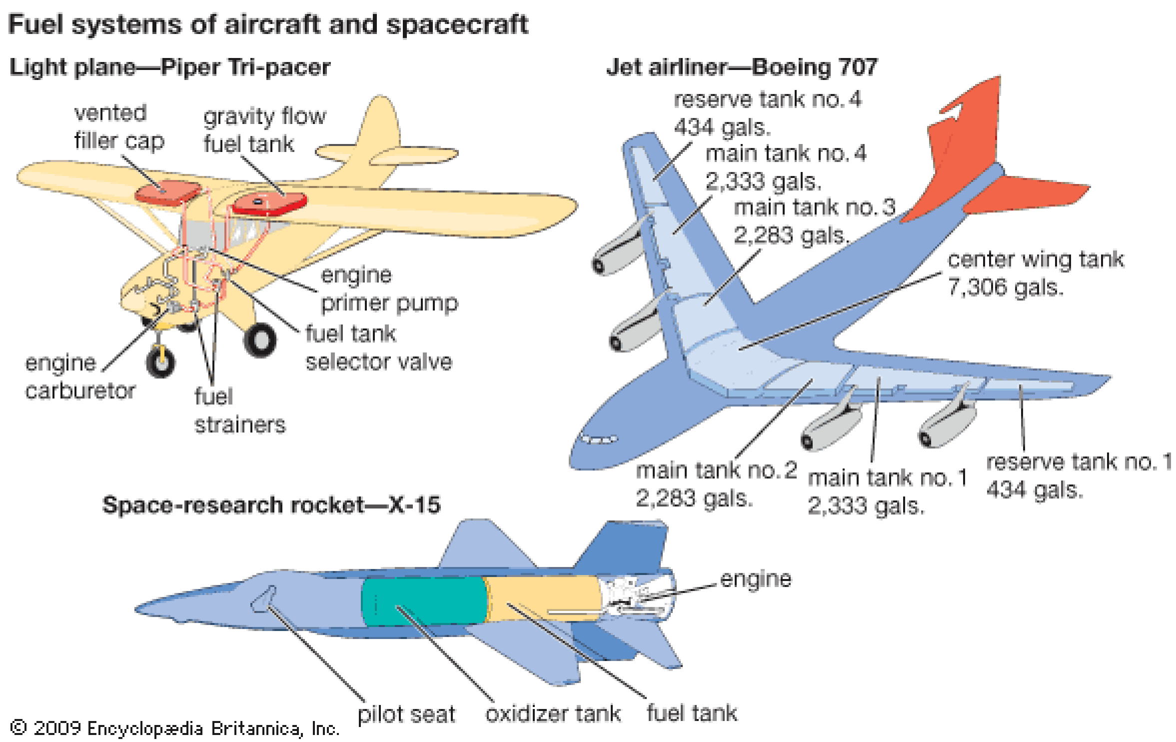
Sizing from a conceptual sketch

The fuel-weight fraction

$$+\left(\frac{W_f}{W_0}\right)$$

- Only part of the aircraft's fuel supply is available for performing the mission ("mission fuel").
- The other fuel includes reserve fuel as required by civil or military design specifications.
- And also includes "trapped fuel," which is the fuel which cannot be pumped out of the tanks.
- The required amount of mission fuel depends upon the mission to be flown, the aerodynamics of the aircraft, and the engine's fuel consumption.

Sizing from a conceptual sketch



Fuel tank locations

Sizing from a conceptual sketch

- Fuel fraction can be estimated based on the mission to be flown using approximations of the fuel consumption and aerodynamics.

- Typical mission profiles for various types of aircraft are shown in Fig. 3.2.

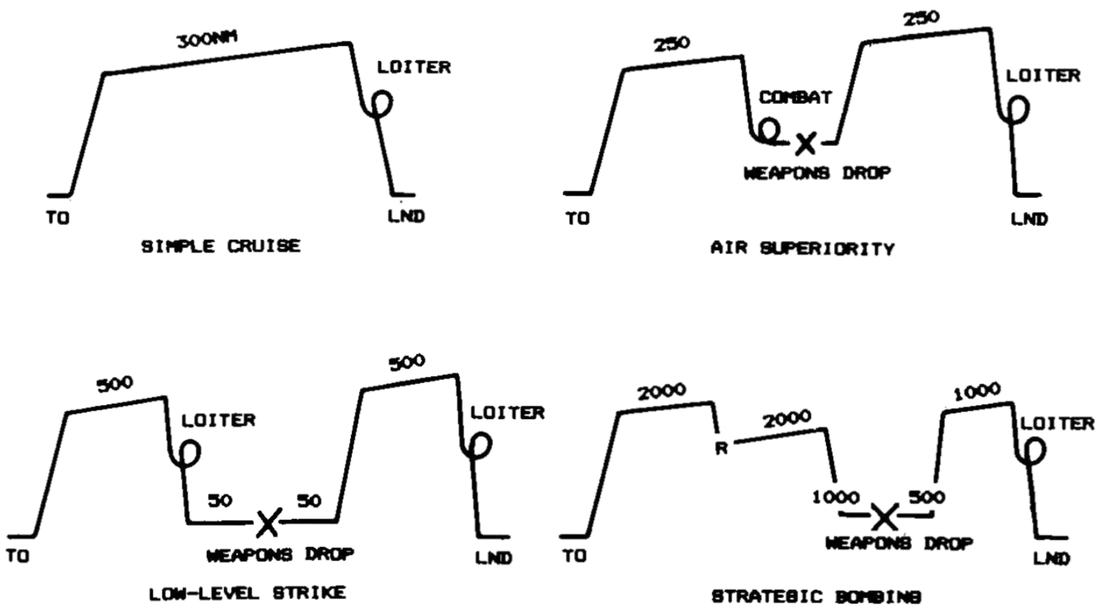


Fig. 3.2 Typical mission profiles for sizing.

Sizing from a conceptual sketch

- The Simple Cruise mission is used for many transport and general aviation designs, including homebuilts.
- The aircraft is sized to provide some required cruise range.
- For safety you would be wise to carry extra fuel in case your intended airport is closed, so a loiter of typically 20-30 min is added.
- Alternatively, additional range could be included, representing the distance to the nearest other airport or some fixed number of minutes of flight at cruise speed.
- FAA requires 30 min of additional cruise fuel for general-aviation aircraft.